

# Updated AR Test Results

May 14, 2026

## Overview

- ▶ **Instrument** An updated version of the instrument was implemented (muni-relative owner-share, pre-earliest-election window).
- ▶ **Two taxonomies.** Aggregate BNDES industries (Agriculture, Industry, Infrastructure, Services); firm-size (Small  $\leq 49$ , Medium 50–249, Big  $\geq 250$ ).
- ▶ **Four control specs.** (1) No controls; (2) + EC (per-cell exposure control); (3) + Vol (volume control); (4) + Vol + EC.
- ▶ Volume control: total BNDES disbursements over 2002 municipal GDP.
- ▶ **Four alignment channels.** Mayor; Mayor · President; Mayor · Governor; Mayor · Gov. · President.
- ▶ **Sample.** 5,544 municipalities; 88,694 municipality-years.
- ▶ All regressions use municipality and year fixed effects; standard errors clustered by municipality.

## Summary of Results

- ▶ The AR test *rejects* at 5% in some specs for both BNDES sectors and firm size.
- ▶ The volume control shifts  $F$  negligibly and does not have a significant nor relevant coefficient; the EC controls (per-cell) move  $F$  in both directions.

## BNDES Sectors — AR joint $F$

Muni-relative owner-share weights, pre-earliest-election window.

| Channel   | No controls    | + EC             | + Vol          | + Vol + EC       |
|-----------|----------------|------------------|----------------|------------------|
| Mayor     | 0.617 [0.604]  | 0.477 [0.699]    | 0.617 [0.604]  | 0.477 [0.699]    |
| M · P     | 1.861 [0.134]  | 1.394 [0.243]    | 1.861 [0.134]  | 1.394 [0.243]    |
| M · G     | 2.147* [0.092] | 3.986*** [0.008] | 2.151* [0.092] | 3.993*** [0.007] |
| M · G · P | 1.022 [0.382]  | 1.126 [0.337]    | 1.022 [0.382]  | 1.127 [0.337]    |

Cluster-robust AR Wald joint  $F$ . \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .  $[p]$  is the  $p$ -value; reject at 5% iff  $p < 0.05$ .

## BNDES Sectors — Coefficients: Mayor

|                | No controls       | + EC              | + Vol             | + Vol + EC        |
|----------------|-------------------|-------------------|-------------------|-------------------|
| Agriculture    | -0.033<br>(0.025) | -0.037<br>(0.032) | -0.033<br>(0.025) | -0.037<br>(0.032) |
| Industry       | 0.003<br>(0.014)  | -0.002<br>(0.015) | 0.003<br>(0.014)  | -0.002<br>(0.015) |
| Infrastructure | 0.002<br>(0.008)  | -0.002<br>(0.010) | 0.002<br>(0.008)  | -0.002<br>(0.010) |
| Volume         | —                 | —                 | 0.000<br>(0.000)  | 0.000<br>(0.000)  |
| EC (mean)      | —                 | 0.009             | —                 | 0.009             |

SE in parentheses. EC (mean) row reports the mean coefficient across the  $K - 1$  EC controls. Stars:

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## BNDES Sectors — Coefficients: Mayor · President

|                | No controls         | + EC               | + Vol               | + Vol + EC         |
|----------------|---------------------|--------------------|---------------------|--------------------|
| Agriculture    | 0.018<br>(0.048)    | 0.036<br>(0.050)   | 0.018<br>(0.048)    | 0.036<br>(0.050)   |
| Industry       | -0.023<br>(0.026)   | -0.011<br>(0.025)  | -0.023<br>(0.026)   | -0.011<br>(0.025)  |
| Infrastructure | -0.044**<br>(0.022) | -0.039*<br>(0.021) | -0.044**<br>(0.022) | -0.039*<br>(0.021) |
| Volume         | —                   | —                  | 0.000<br>(0.000)    | 0.000<br>(0.000)   |
| EC (mean)      | —                   | -0.019             | —                   | -0.019             |

SE in parentheses. EC (mean) row reports the mean coefficient across the  $K - 1$  EC controls.

## BNDES Sectors — Coefficients: Mayor · Governor

|                | No controls        | + EC                | + Vol              | + Vol + EC          |
|----------------|--------------------|---------------------|--------------------|---------------------|
| Agriculture    | -0.032<br>(0.035)  | -0.025<br>(0.037)   | -0.032<br>(0.035)  | -0.025<br>(0.037)   |
| Industry       | 0.012<br>(0.019)   | 0.022<br>(0.020)    | 0.012<br>(0.019)   | 0.022<br>(0.020)    |
| Infrastructure | 0.021**<br>(0.010) | 0.030***<br>(0.010) | 0.021**<br>(0.010) | 0.030***<br>(0.010) |
| Volume         | —                  | —                   | 0.000<br>(0.000)   | 0.000<br>(0.000)    |
| EC (mean)      | —                  | -0.021              | —                  | -0.021              |

SE in parentheses. EC (mean) row reports the mean coefficient across the  $K - 1$  EC controls.

## BNDES Sectors — Coefficients: Mayor · Gov. · President

|                | No controls       | + EC              | + Vol             | + Vol + EC        |
|----------------|-------------------|-------------------|-------------------|-------------------|
| Agriculture    | 0.082<br>(0.056)  | 0.101*<br>(0.058) | 0.082<br>(0.056)  | 0.101*<br>(0.058) |
| Industry       | -0.021<br>(0.033) | -0.010<br>(0.032) | -0.021<br>(0.033) | -0.010<br>(0.032) |
| Infrastructure | -0.026<br>(0.037) | -0.020<br>(0.037) | -0.026<br>(0.037) | -0.020<br>(0.037) |
| Volume         | —                 | —                 | 0.000<br>(0.000)  | 0.000<br>(0.000)  |
| EC (mean)      | —                 | -0.020            | —                 | -0.020            |

SE in parentheses. EC (mean) row reports the mean coefficient across the  $K - 1$  EC controls.



## Firm Size — AR joint $F$

Muni-relative owner-share weights, pre-earliest-election window.

| Channel   | No controls     | + EC            | + Vol           | + Vol + EC      |
|-----------|-----------------|-----------------|-----------------|-----------------|
| Mayor     | 0.282 [0.755]   | 0.144 [0.866]   | 0.282 [0.755]   | 0.144 [0.866]   |
| M · P     | 4.156** [0.016] | 1.649 [0.192]   | 4.156** [0.016] | 1.649 [0.192]   |
| M · G     | 0.574 [0.563]   | 4.069** [0.017] | 0.574 [0.563]   | 4.065** [0.017] |
| M · G · P | 1.267 [0.282]   | 0.597 [0.551]   | 1.267 [0.282]   | 0.597 [0.551]   |

Cluster-robust AR Wald joint  $F$ . \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . [ $p$ ] is the  $p$ -value; reject at 5% iff  $p < 0.05$ .

## Firm Size — Coefficients: Mayor

|           | No controls       | + EC              | + Vol             | + Vol + EC        |
|-----------|-------------------|-------------------|-------------------|-------------------|
| Small     | -0.003<br>(0.006) | 0.003<br>(0.007)  | -0.003<br>(0.006) | 0.003<br>(0.007)  |
| Medium    | -0.004<br>(0.007) | -0.003<br>(0.008) | -0.004<br>(0.007) | -0.003<br>(0.008) |
| Volume    | —                 | —                 | 0.000<br>(0.000)  | 0.000<br>(0.000)  |
| EC (mean) | —                 | -0.010            | —                 | -0.010            |

SE in parentheses. EC (mean) row reports the mean coefficient across the  $K - 1$  EC controls. Stars:

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## Firm Size — Coefficients: Mayor · President

|           | No controls          | + EC               | + Vol                | + Vol + EC         |
|-----------|----------------------|--------------------|----------------------|--------------------|
| Small     | -0.033***<br>(0.011) | -0.020*<br>(0.011) | -0.033***<br>(0.011) | -0.020*<br>(0.011) |
| Medium    | 0.003<br>(0.013)     | 0.005<br>(0.014)   | 0.003<br>(0.013)     | 0.005<br>(0.014)   |
| Volume    | —                    | —                  | 0.000<br>(0.000)     | 0.000<br>(0.000)   |
| EC (mean) | —                    | -0.028             | —                    | -0.028             |

SE in parentheses. EC (mean) row reports the mean coefficient across the  $K - 1$  EC controls.

## Firm Size — Coefficients: Mayor · Governor

|           | No controls      | + EC                | + Vol            | + Vol + EC          |
|-----------|------------------|---------------------|------------------|---------------------|
| Small     | 0.006<br>(0.008) | 0.023***<br>(0.009) | 0.006<br>(0.008) | 0.023***<br>(0.009) |
| Medium    | 0.007<br>(0.010) | 0.008<br>(0.011)    | 0.007<br>(0.010) | 0.008<br>(0.011)    |
| Volume    | —                | —                   | 0.000<br>(0.000) | 0.000<br>(0.000)    |
| EC (mean) | —                | -0.032              | —                | -0.032              |

SE in parentheses. EC (mean) row reports the mean coefficient across the  $K - 1$  EC controls.

## Firm Size — Coefficients: Mayor · Gov. · President

|           | No controls       | + EC              | + Vol             | + Vol + EC        |
|-----------|-------------------|-------------------|-------------------|-------------------|
| Small     | -0.024<br>(0.017) | -0.012<br>(0.017) | -0.024<br>(0.017) | -0.012<br>(0.017) |
| Medium    | 0.018<br>(0.019)  | 0.017<br>(0.019)  | 0.018<br>(0.019)  | 0.017<br>(0.019)  |
| Volume    | —                 | —                 | 0.000<br>(0.000)  | 0.000<br>(0.000)  |
| EC (mean) | —                 | -0.029            | —                 | -0.029            |

SE in parentheses. EC (mean) row reports the mean coefficient across the  $K - 1$  EC controls.